

# Exact Trigonometric Table of Integral Angles

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## Abstract

This document gives a compact exact reference for  $\sin A^\circ$  and  $\cos A^\circ$  for every integer angle  $A$  from  $0^\circ$  through  $45^\circ$ . The expressions are generated constructively from a finite radical anchor set using the standard trigonometric identities below. No numerical approximation or post-processing by RootReduce or ToRadicals is required: the quantities are stored directly in exact radical form.

The document ends with “The End”

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# 1 Trigonometric Identities

## 1.1 Pythagorean Identity

For every angle  $\theta$ ,

$$\sin^2 \theta + \cos^2 \theta = 1.$$

## 1.2 Addition Formulae

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta,$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta.$$

## 1.3 Subtraction Formulae

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta,$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta.$$

## 1.4 Double-Angle Formulae

$$\sin 2\theta = 2 \sin \theta \cos \theta, \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta,$$

$$\sin \theta = \sqrt{\frac{1 - \cos 2\theta}{2}}, \quad \cos \theta = \sqrt{\frac{1 + \cos 2\theta}{2}}.$$

## 1.5 Triple-Angle Formulae

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta, \quad \cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta.$$

## 1.6 Complementary and Quadrant Identities

$$\sin(90^\circ - \theta) = \cos \theta, \quad \cos(90^\circ - \theta) = \sin \theta.$$

For the complete circle,

$$90^\circ \leq A \leq 180^\circ : \quad \sin A = \sin(180^\circ - A), \quad \cos A = -\cos(180^\circ - A),$$

$$180^\circ \leq A \leq 270^\circ : \quad \sin A = -\sin(A - 180^\circ), \quad \cos A = -\cos(A - 180^\circ),$$

$$270^\circ \leq A \leq 360^\circ : \quad \sin A = -\sin(360^\circ - A), \quad \cos A = \cos(360^\circ - A).$$

## 2 Radical Anchor Set

Define  $S_A = \sin A^\circ$  and  $C_A = \cos A^\circ$ . The fundamental exact generators are

$$\begin{aligned}
 S_0 &= 0, & C_0 &= 1, \\
 S_{30} &= \frac{1}{2}, & C_{30} &= \frac{\sqrt{3}}{2}, \\
 S_{45} &= C_{45} = \frac{\sqrt{2}}{2}, \\
 S_{60} &= \frac{\sqrt{3}}{2}, & C_{60} &= \frac{1}{2}, \\
 S_{90} &= 1, & C_{90} &= 0, \\
 S_{15} &= \frac{\sqrt{6} - \sqrt{2}}{4}, & C_{15} &= \frac{\sqrt{6} + \sqrt{2}}{4}, \\
 S_{18} &= \frac{\sqrt{5} - 1}{4}, & C_{18} &= \frac{\sqrt{10 + 2\sqrt{5}}}{4}.
 \end{aligned}$$

For  $20^\circ$ , let

$$R = \frac{\sqrt[3]{2 + \sqrt{3}} + \sqrt[3]{2 - \sqrt{3}}}{2} = C_{20},$$

so that

$$S_{20} = \sqrt{1 - R^2}.$$

The remaining fundamental values are constructed by

$$\begin{aligned}
 S_{10} &= \sqrt{\frac{1 - R}{2}}, & C_{10} &= \sqrt{\frac{1 + R}{2}}, \\
 S_5 &= \sqrt{\frac{1 - C_{10}}{2}}, & C_5 &= \sqrt{\frac{1 + C_{10}}{2}}, \\
 S_9 &= \sqrt{\frac{1 - C_{18}}{2}}, & C_9 &= \sqrt{\frac{1 + C_{18}}{2}}, \\
 S_4 &= S_9 C_5 - C_9 S_5, & C_4 &= C_9 C_5 + S_9 S_5, \\
 S_2 &= \sqrt{\frac{1 - C_4}{2}}, & C_2 &= \sqrt{\frac{1 + C_4}{2}}, \\
 S_1 &= \sqrt{\frac{1 - C_2}{2}}, & C_1 &= \sqrt{\frac{1 + C_2}{2}}, \\
 S_3 &= S_{18} C_{15} - C_{18} S_{15}, & C_3 &= C_{18} C_{15} + S_{18} S_{15}.
 \end{aligned}$$

Thus every  $S_k, C_k$  used below is an exact radical expression. Additional

nodes are

$$\begin{aligned}
S_6 &= S_3 C_3 + C_3 S_3, & C_6 &= C_3^2 - S_3^2, \\
S_7 &= S_5 C_2 + C_5 S_2, & C_7 &= C_5 C_2 - S_5 S_2, \\
S_8 &= S_4 C_4 + C_4 S_4, & C_8 &= C_4^2 - S_4^2, \\
S_{36} &= 2S_{18} C_{18}, & C_{36} &= C_{18}^2 - S_{18}^2, \\
S_{54} &= 3S_{18} - 4S_{18}^3, & C_{54} &= 4C_{18}^3 - 3C_{18}, \\
S_{75} &= S_{45} C_{30} + C_{45} S_{30}, & C_{75} &= C_{45} C_{30} - S_{45} S_{30}.
\end{aligned}$$

### 3 $\sin A$ and $\cos A$

$$A = 0^\circ$$

$$\boxed{\sin 0^\circ = 0}.$$

$$\boxed{\cos 0^\circ = 1}.$$

$$A = 1^\circ$$

$$\boxed{\sin 1^\circ = S_1}.$$

$$\boxed{\cos 1^\circ = C_1}.$$

$$A = 2^\circ$$

$$\boxed{\sin 2^\circ = S_2}.$$

$$\boxed{\cos 2^\circ = C_2}.$$

$$A = 3^\circ$$

$$\boxed{\sin 3^\circ = S_{18} C_{15} - C_{18} S_{15}}.$$

$$\boxed{\cos 3^\circ = C_{18} C_{15} + S_{18} S_{15}}.$$

$$A = 4^\circ$$

$$\boxed{\sin 4^\circ = S_9 C_5 - C_9 S_5}.$$

$$\boxed{\cos 4^\circ = C_9 C_5 + S_9 S_5}.$$

$$A = 5^\circ$$

$$\sin 5^\circ = S_5.$$

$$\cos 5^\circ = C_5.$$

$$A = 6^\circ$$

$$\sin 6^\circ = S_3 C_3 + C_3 S_3.$$

$$\cos 6^\circ = C_3^2 - S_3^2.$$

$$A = 7^\circ$$

$$\sin 7^\circ = S_4 C_3 + C_4 S_3.$$

$$\cos 7^\circ = C_4 C_3 - S_4 S_3.$$

$$A = 8^\circ$$

$$\sin 8^\circ = 2S_4 C_4.$$

$$\cos 8^\circ = C_4^2 - S_4^2.$$

$$A = 9^\circ$$

$$\sin 9^\circ = S_9 = \sqrt{\frac{1 - C_{18}}{2}}.$$

$$\cos 9^\circ = C_9 = \sqrt{\frac{1 + C_{18}}{2}}.$$

$$A = 10^\circ$$

$$\sin 10^\circ = S_{10} = \sqrt{\frac{1 - R}{2}}.$$

$$\cos 10^\circ = C_{10} = \sqrt{\frac{1 + R}{2}}.$$

$$A = 11^\circ$$

$$\sin 11^\circ = S_2 C_9 + C_2 S_9.$$

$$\cos 11^\circ = C_2 C_9 - S_2 S_9.$$

$$A = 12^\circ$$

$$\sin 12^\circ = S_3 C_9 + C_3 S_9.$$

$$\cos 12^\circ = C_3 C_9 - S_3 S_9.$$

$$A = 13^\circ$$

$$\sin 13^\circ = S_9 C_4 + C_9 S_4.$$

$$\cos 13^\circ = C_9 C_4 - S_9 S_4.$$

$$A = 14^\circ$$

$$\sin 14^\circ = S_9 C_5 + C_9 S_5.$$

$$\cos 14^\circ = C_9 C_5 - S_9 S_5.$$

$$A = 15^\circ$$

$$\sin 15^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}.$$

$$\cos 15^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

$$A = 16^\circ$$

$$\sin 16^\circ = S_{15} C_1 + C_{15} S_1.$$

$$\cos 16^\circ = C_{15} C_1 - S_{15} S_1.$$

$$A = 17^\circ$$

$$\sin 17^\circ = S_{15} C_2 + C_{15} S_2.$$

$$\cos 17^\circ = C_{15} C_2 - S_{15} S_2.$$

$$A = 18^\circ$$

$$\sin 18^\circ = \frac{\sqrt{5} - 1}{4}.$$

$$\cos 18^\circ = \frac{\sqrt{10 + 2\sqrt{5}}}{4}.$$

$$A = 19^\circ$$

$$\sin 19^\circ = S_{10}C_9 + C_{10}S_9.$$

$$\cos 19^\circ = C_{10}C_9 - S_{10}S_9.$$

$$A = 20^\circ$$

$$\sin 20^\circ = \sqrt{1 - R^2}.$$

$$\cos 20^\circ = R.$$

$$A = 21^\circ$$

$$\sin 21^\circ = S_{18}C_3 + C_{18}S_3.$$

$$\cos 21^\circ = C_{18}C_3 - S_{18}S_3.$$

$$A = 22^\circ$$

$$\sin 22^\circ = S_{18}C_4 + C_{18}S_4.$$

$$\cos 22^\circ = C_{18}C_4 - S_{18}S_4.$$

$$A = 23^\circ$$

$$\sin 23^\circ = S_{18}C_5 + C_{18}S_5.$$

$$\cos 23^\circ = C_{18}C_5 - S_{18}S_5.$$

$$A = 24^\circ$$

$$\sin 24^\circ = S_{15}C_9 + C_{15}S_9.$$

$$\cos 24^\circ = C_{15}C_9 - S_{15}S_9.$$

$$A = 25^\circ$$

$$\sin 25^\circ = S_{15}C_{10} + C_{15}S_{10}.$$

$$\cos 25^\circ = C_{15}C_{10} - S_{15}S_{10}.$$

$$A = 26^\circ$$

$$\sin 26^\circ = S_{30}C_4 - C_{30}S_4.$$

$$\cos 26^\circ = C_{30}C_4 + S_{30}S_4.$$

$$A = 27^\circ$$

$$\sin 27^\circ = S_{18}C_9 + C_{18}S_9.$$

$$\cos 27^\circ = C_{18}C_9 - S_{18}S_9.$$

$$A = 28^\circ$$

$$\sin 28^\circ = S_{18}C_{10} + C_{18}S_{10}.$$

$$\cos 28^\circ = C_{18}C_{10} - S_{18}S_{10}.$$

$$A = 29^\circ$$

$$\sin 29^\circ = S_{20}C_9 + C_{20}S_9.$$

$$\cos 29^\circ = C_{20}C_9 - S_{20}S_9.$$

$$A = 30^\circ$$

$$\sin 30^\circ = \frac{1}{2}.$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}.$$



$$A = 31^\circ$$

$$\sin 31^\circ = S_{30}C_1 + C_{30}S_1.$$

$$\cos 31^\circ = C_{30}C_1 - S_{30}S_1.$$

$$A = 32^\circ$$

$$\sin 32^\circ = S_{30}C_2 + C_{30}S_2.$$

$$\cos 32^\circ = C_{30}C_2 - S_{30}S_2.$$

$$A = 33^\circ$$

$$\sin 33^\circ = S_{18}C_{15} + C_{18}S_{15}.$$

$$\cos 33^\circ = C_{18}C_{15} - S_{18}S_{15}.$$

$$A = 34^\circ$$

$$\sin 34^\circ = S_{30}C_4 + C_{30}S_4.$$

$$\cos 34^\circ = C_{30}C_4 - S_{30}S_4.$$

$$A = 35^\circ$$

$$\sin 35^\circ = S_{20}C_{15} + C_{20}S_{15}.$$

$$\cos 35^\circ = C_{20}C_{15} - S_{20}S_{15}.$$

$$A = 36^\circ$$

$$\sin 36^\circ = 2S_{18}C_{18}.$$

$$\cos 36^\circ = C_{18}^2 - S_{18}^2.$$

$$A = 37^\circ$$

$$\sin 37^\circ = S_{45}C_8 - C_{45}S_8.$$

$$\cos 37^\circ = C_{45}C_8 + S_{45}S_8.$$

$$A = 38^\circ$$

$$\sin 38^\circ = S_{20}C_{18} + C_{20}S_{18}.$$

$$\cos 38^\circ = C_{20}C_{18} - S_{20}S_{18}.$$

$$A = 39^\circ$$

$$\sin 39^\circ = S_{30}C_9 + C_{30}S_9.$$

$$\cos 39^\circ = C_{30}C_9 - S_{30}S_9.$$

$$A = 40^\circ$$

$$\sin 40^\circ = 2S_{20}C_{20}.$$

$$\cos 40^\circ = C_{20}^2 - S_{20}^2.$$

$$A = 41^\circ$$

$$\sin 41^\circ = S_{45}C_4 - C_{45}S_4.$$

$$\cos 41^\circ = C_{45}C_4 + S_{45}S_4.$$

$$A = 42^\circ$$

$$\sin 42^\circ = S_{45}C_3 - C_{45}S_3.$$

$$\cos 42^\circ = C_{45}C_3 + S_{45}S_3.$$

$$A = 43^\circ$$

$$\sin 43^\circ = S_{45}C_2 - C_{45}S_2.$$

$$\cos 43^\circ = C_{45}C_2 + S_{45}S_2.$$

$$A = 44^\circ$$

$$\sin 44^\circ = S_{45}C_1 - C_{45}S_1.$$

$$\cos 44^\circ = C_{45}C_1 + S_{45}S_1.$$

$$A = 45^\circ$$

$$\sin 45^\circ = \frac{\sqrt{2}}{2}.$$

$$\cos 45^\circ = \frac{\sqrt{2}}{2}.$$

## 4 Constructive Completeness

The preceding sections reproduce the constructive first-quadrant table from  $0^\circ$  through  $45^\circ$ . Every displayed  $S_k$  and  $C_k$  is a radical expression by the anchor definitions and the closure of radicals under rational arithmetic, square roots, and the conjugate cubic radicals entering  $R = C_{20}$ . Consequently every displayed sine and cosine value is an exact radical expression.

The same construction extends to the complete circle through complementary-angle and quadrant symmetries. Thus every integral angle in  $0^\circ \leq A \leq 360^\circ$  is generated by a finite sequence of exact algebraic operations from the fundamental radical anchors.

**The End**